

Telco Customer Churn - Executive Overview

Historical Patterns | Customer Risk | Model Trade Off

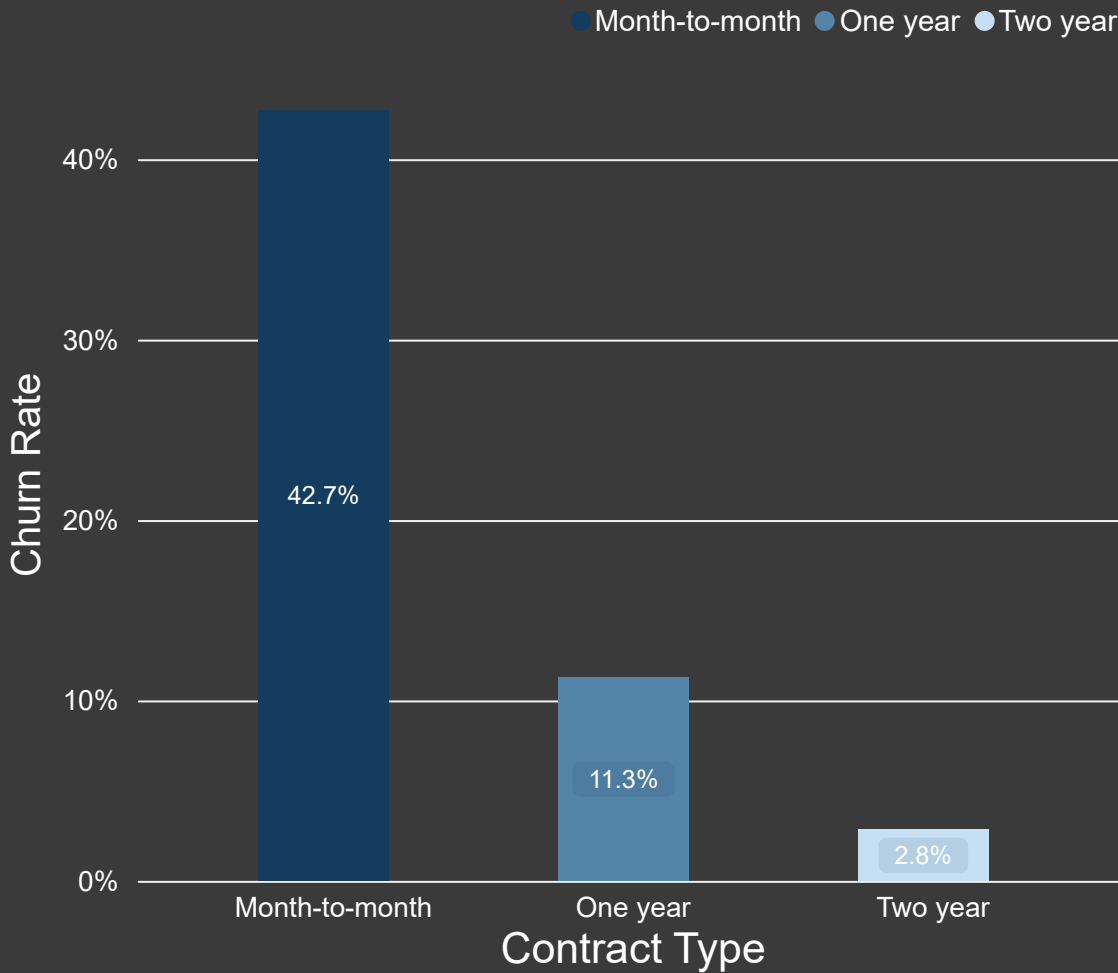
High Risk Customers
2298

Customers in File
7043

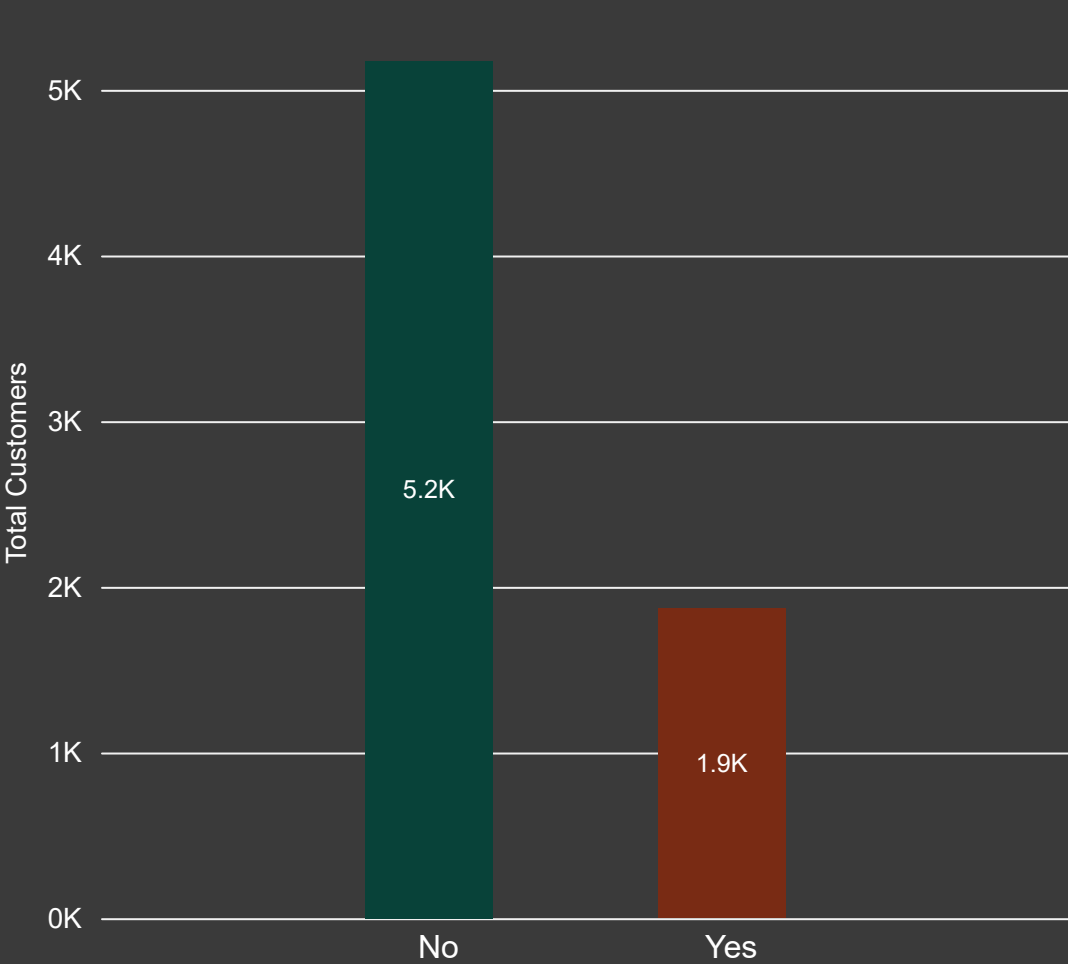
Actual Churn Rate
26.54%

Churned Customers
1869

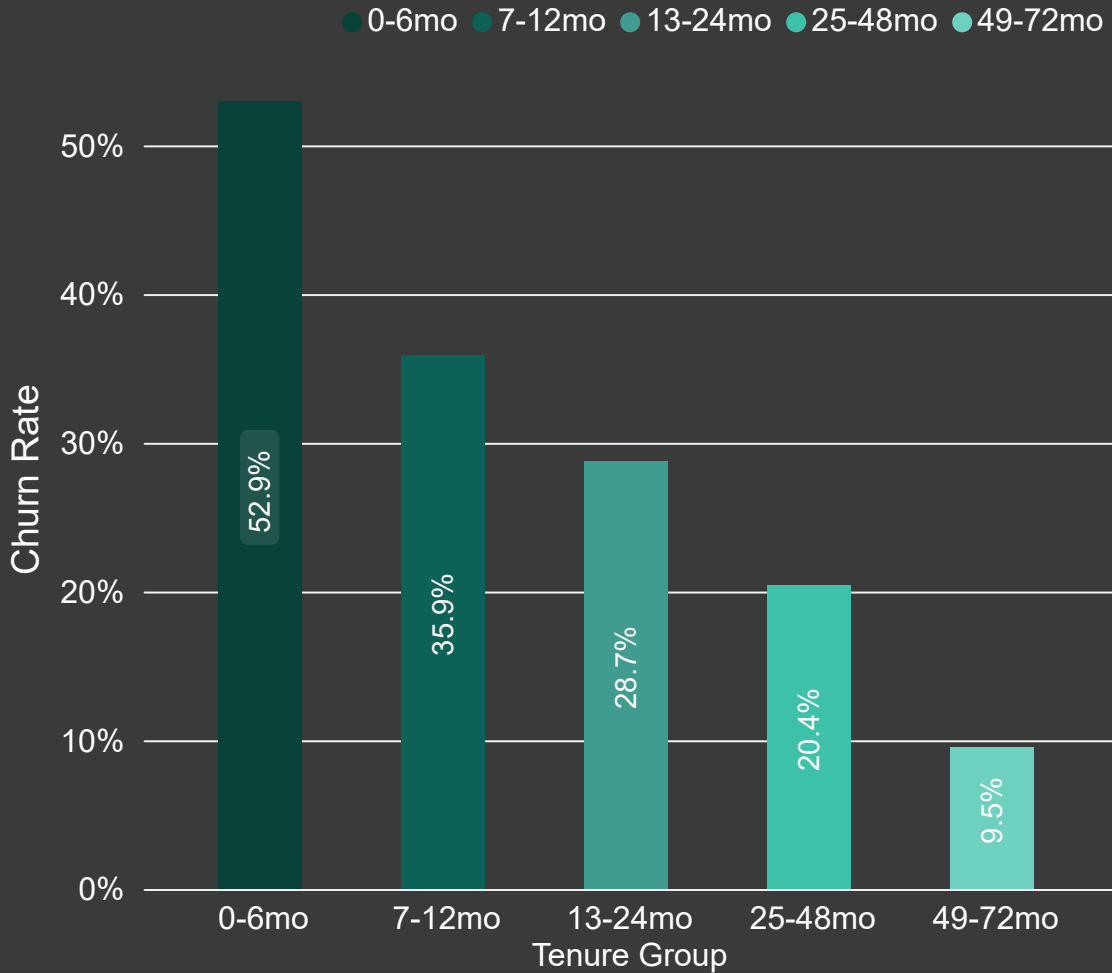
Churn Rate by Contract Type



Total Customers by Churn



Churn Rate by Tenure Group



Historical churn affected 26.5% of customers. Churn is substantially higher among month-to-month customers and customers with shorter tenure, suggesting that retention risk is concentrated rather than evenly distributed across the customer base.

Where Does Churn Vary

Contract

All

InternetService

All

TenureGroup

-

PaymentMethod

All

Contract	DSL	Fiber optic	No	Total
Month-to-month	32.2%	54.6%	18.9%	42.7%
One year	9.3%	19.3%	2.5%	11.3%
Two year	1.9%	7.2%	0.8%	2.8%
Total	19.0%	41.9%	7.4%	26.5%

Contract	0-6mo	13-24mo	25-48mo	49-72mo	7-12mo	Total
Month-to-month	55.2%	37.7%	32.9%	26.0%	42.0%	42.7%
One year	10.3%	8.1%	10.6%	12.9%	10.6%	11.3%
Two year			2.2%	3.3%		2.8%
Total	52.9%	28.7%	20.4%	9.5%	35.9%	26.5%

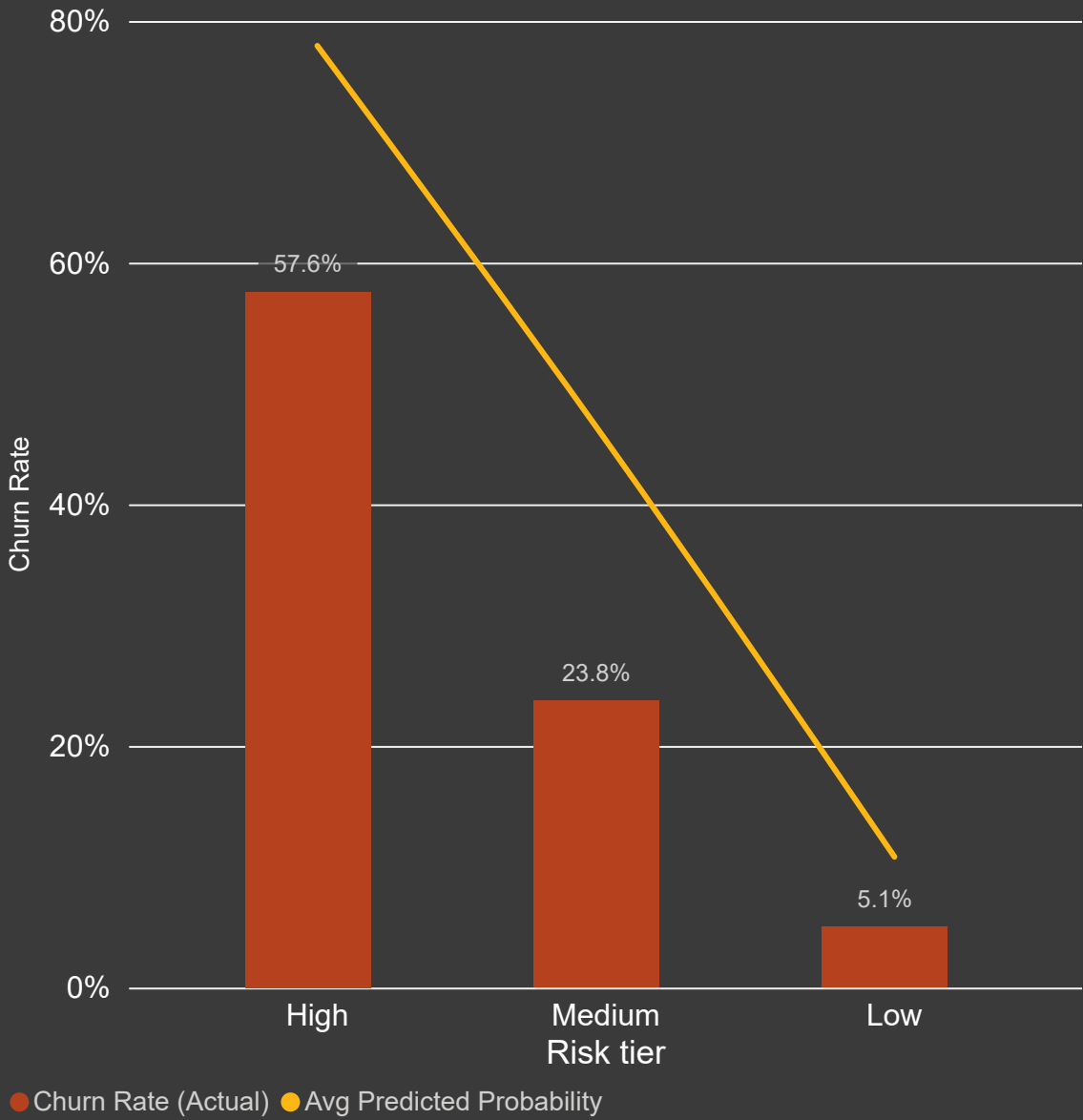
Contract	Bank transfer (automatic)	Credit card (automatic)	Electronic check	Mailed check	Total
Month-to-month	34.1%	32.8%	53.7%	31.6%	42.7%
One year	9.7%	10.3%	18.4%	6.8%	11.3%
Two year	3.4%	2.2%	7.7%	0.8%	2.8%
Total	16.7%	15.2%	45.3%	19.1%	26.5%

Customer Risk Profile

High Risk Rate
32.63%

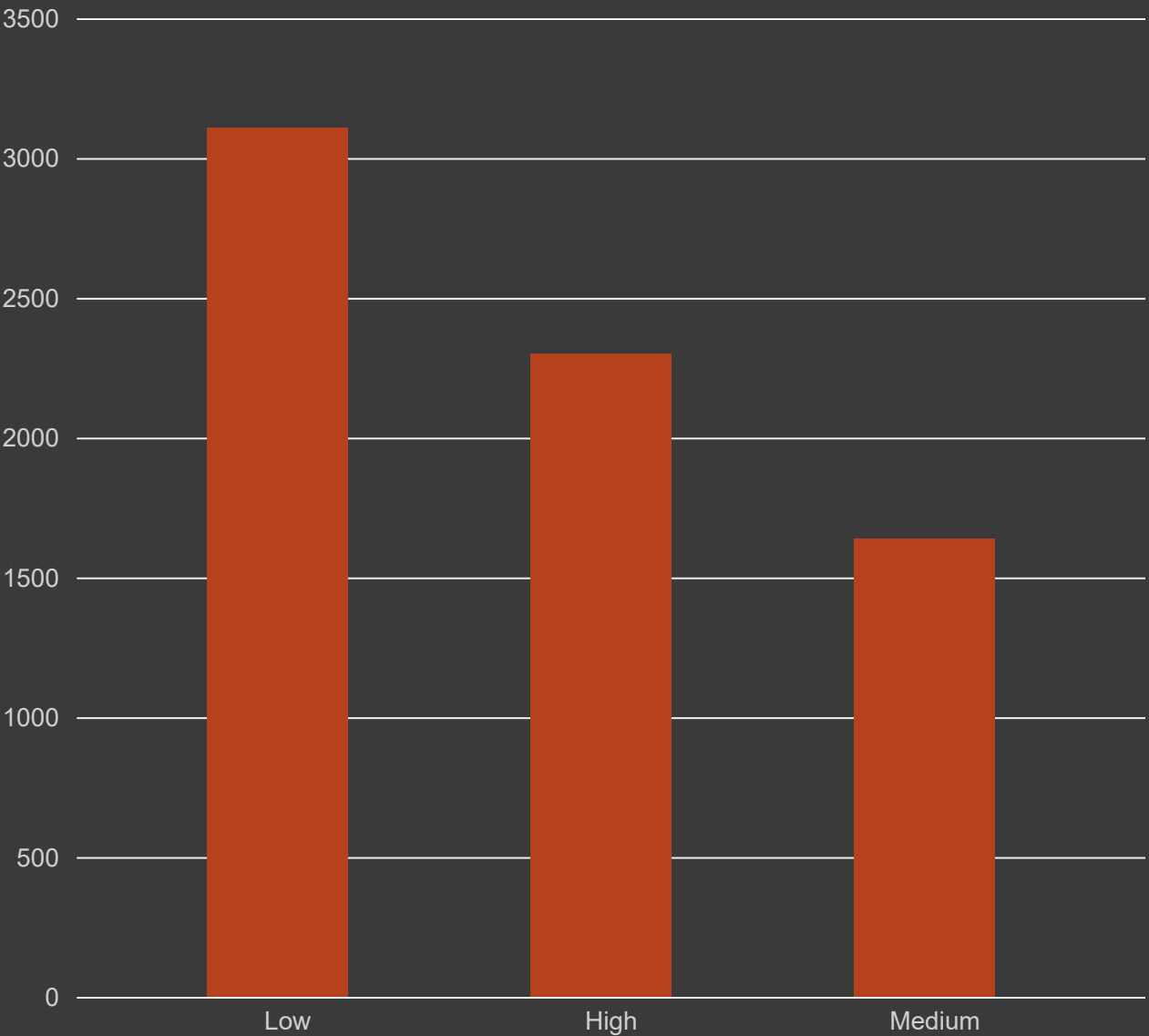
customerID	TenureGroup	Contract	InternetService	Sum of MonthlyCharges	Sum of ChurnProbability	RiskTier
0004-TLHLJ	0-6mo	Month-to-month	Fiber optic	73.90	0.90	High
0011-IGKFF	13-24mo	Month-to-month	Fiber optic	98.00	0.83	High
0013-EXCHZ	0-6mo	Month-to-month	Fiber optic	83.90	0.85	High
0018-NYROU	0-6mo	Month-to-month	Fiber optic	68.95	0.90	High
0021-IKXGC	0-6mo	Month-to-month	Fiber optic	72.10	0.94	High
Total				456,116.60	2,868.90	

Observed Churn vs Predicted Risk by Customer Risk Tier



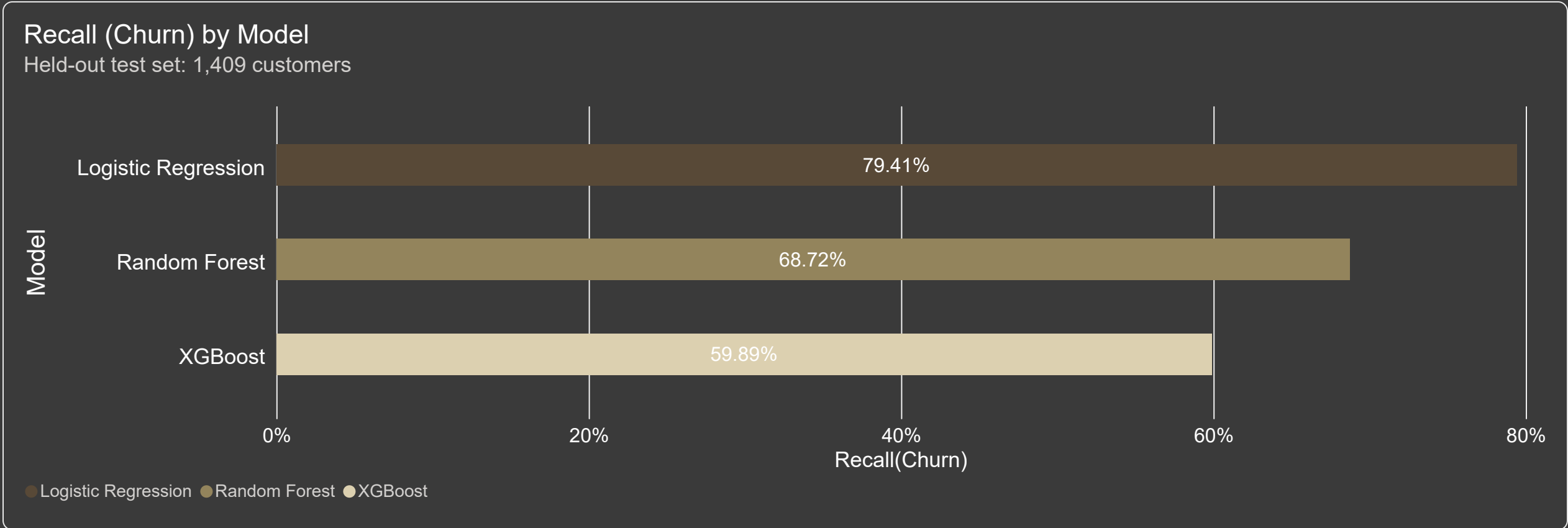
- Risk tiers are based on model-predicted churn probability: Low <30%, Medium 30–<60%, and High ≥60%.
- Actual churn rate represents the historical churn observed within each tier.

Total Customers by RiskTier



Model Performance

Model	Sum of Accuracy	Sum of AUC	Sum of Brier Score	Sum of F1 (Churn)	Precision (Churn)	Sum of Recall (Churn)
Logistic Regression	74.31%	84.26%	16.57%	62.13%	51.03%	79.41%
Random Forest	76.93%	84.11%	15.03%	61.26%	55.27%	68.72%
XGBoost	78.14%	84.30%	13.99%	59.26%	58.64%	59.89%



Why was Logistic Regression Selected?

Logistic Regression achieved the highest recall on the held-out test set at 79.4%. Because the project prioritises identifying as many actual churners as possible, recall was used as the primary model-selection criterion. This does not mean Logistic Regression performs best on every metric; XGBoost, for example, produced higher precision and accuracy.

Why does Changing the Threshold matter?

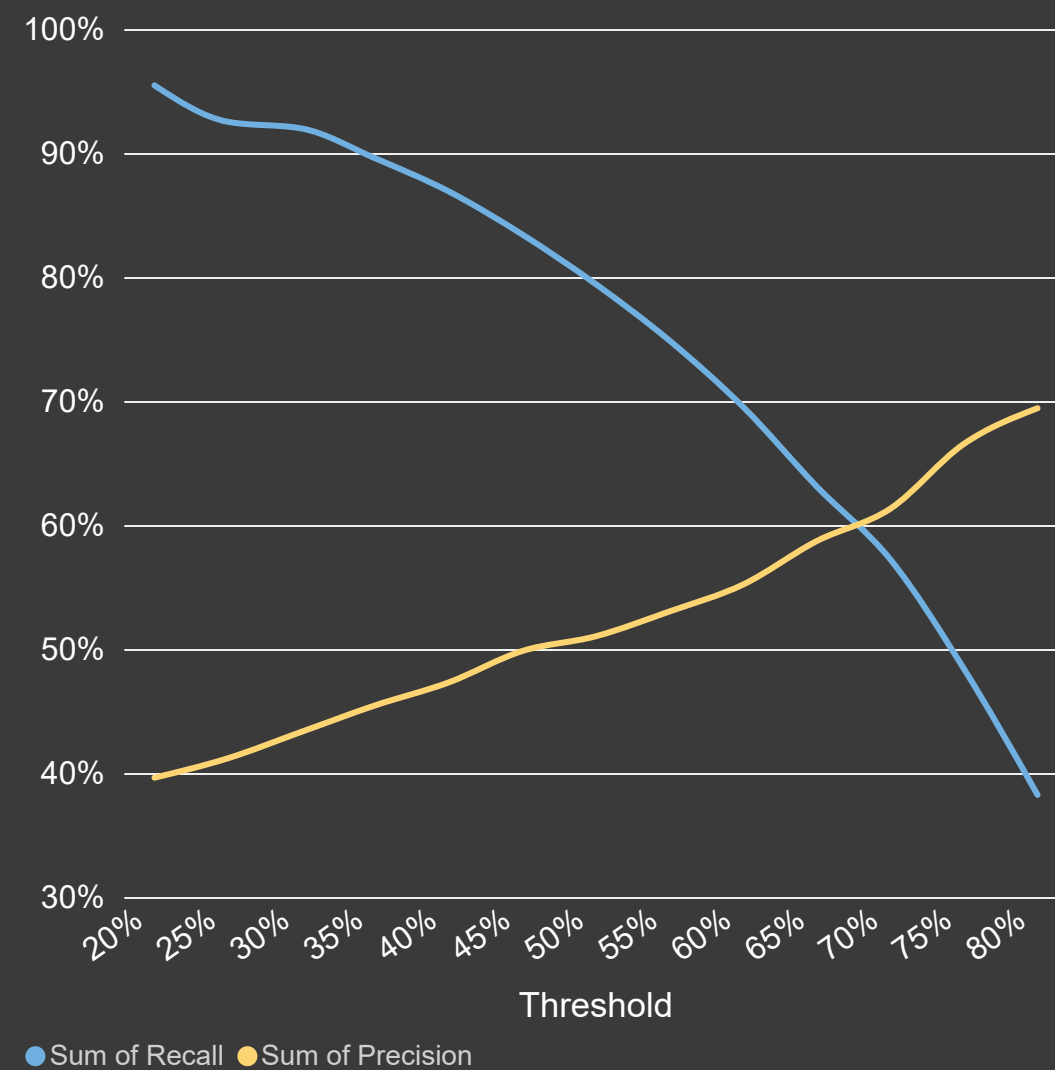
The model produces a probability between 0 and 1. A classification threshold determines when a customer is labelled "predicted churn". Lower thresholds identify more potential churners but also flag more customers who may not churn.

Threshold

20%



Sum of Recall and Sum of Precision
by Threshold



Sum of FlaggedCustomers

901

Recall @ Threshold

95.45%

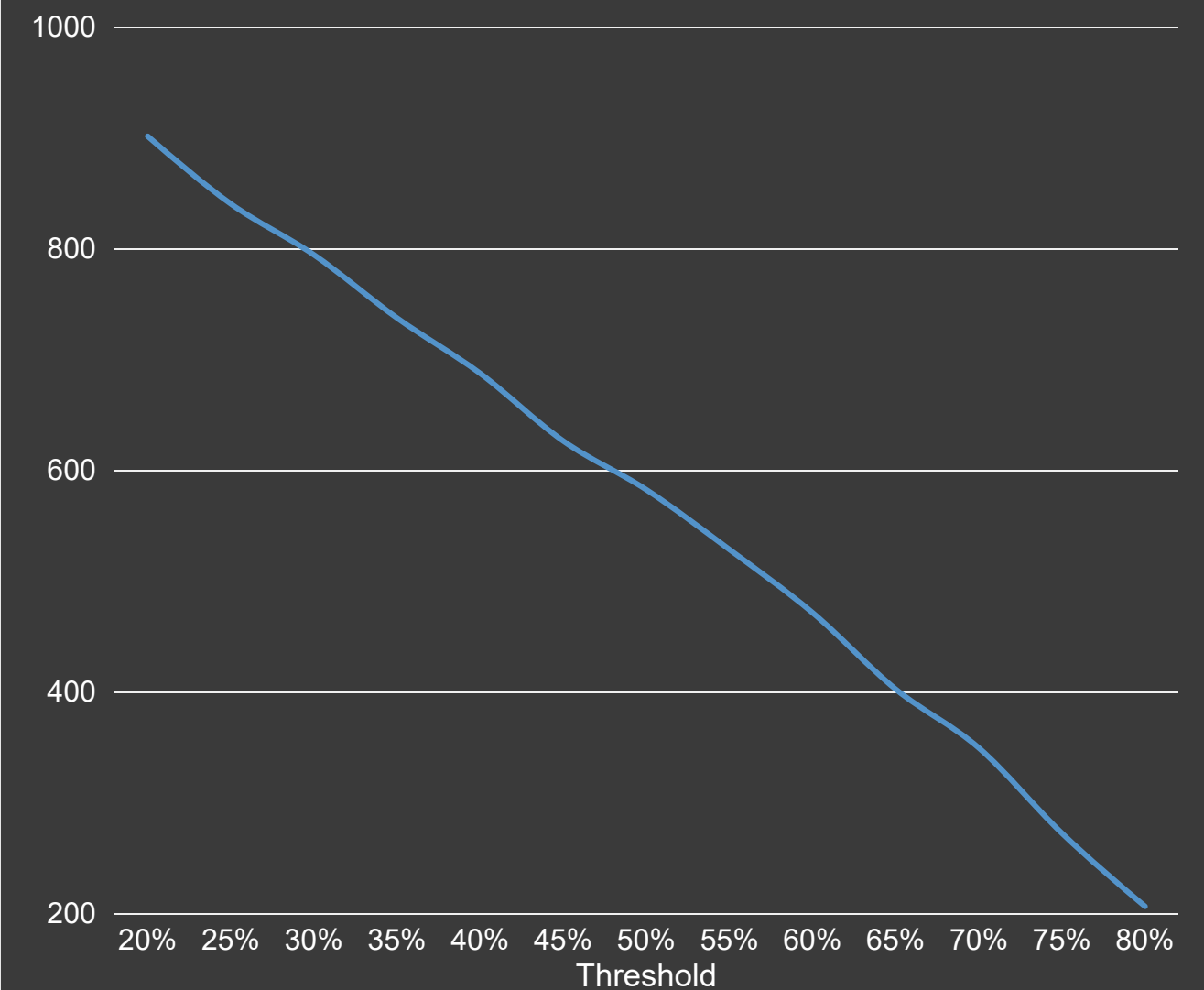
Precision @ Threshold

39.62%

Sum of F1

56.00%

Sum of FlaggedCustomers
by Threshold



Actions, Validation and Limitations

Historical outcomes: 7,043 customers. Model evaluation: 1,409 held-out test customers.

What the Data Shows

52.94%

Customers in their first 6 months churned at 52.94%, against 26.5% overall. This group is 21% of customers but 42% of all churners.

54.61%

Month-to-month customers on Fiber (2,128 customers) churned at 54.61%. They are 30% of customers but 62% of all churners.

70.79%

Of all churners, 70.79% sit in the High risk tier, which is 33% of customers. Observed churn rises from 5% to 24% to 58% across the Low, Medium and High tiers (all 7,043 customers scored).

What to do next

Review early-tenure onboarding. Investigate what is different about the first 6 months, such as setup, early support contacts and first bills.

Examine the month-to-month Fiber segment. Compare pricing, service quality and contract options with other segments to see what drives the gap.

Pilot a retention offer with a holdout group. Target High-tier month-to-month customers, size the list to retention-team capacity, and measure the churn difference before scaling.

What to keep in mind

Association, not causation. These patterns describe past behaviour. They don't show why customers leave or that an offer will keep them.

Scores are rankings, not literal probabilities. The model was trained on SMOTE-balanced data, so predictions run higher than reality (40.7% average predicted vs 26.5% actual churn).

Risk tiers are partly in-sample. All 7,043 customers were scored by a model trained on 80% of them. Model metrics use only the 1,409 test customers.

Model and threshold choice need judgement. Logistic Regression led on recall at a 0.5 cutoff, but AUC is nearly identical across models (0.841 to 0.843). The right cutoff depends on contact capacity and the cost of each type of error.